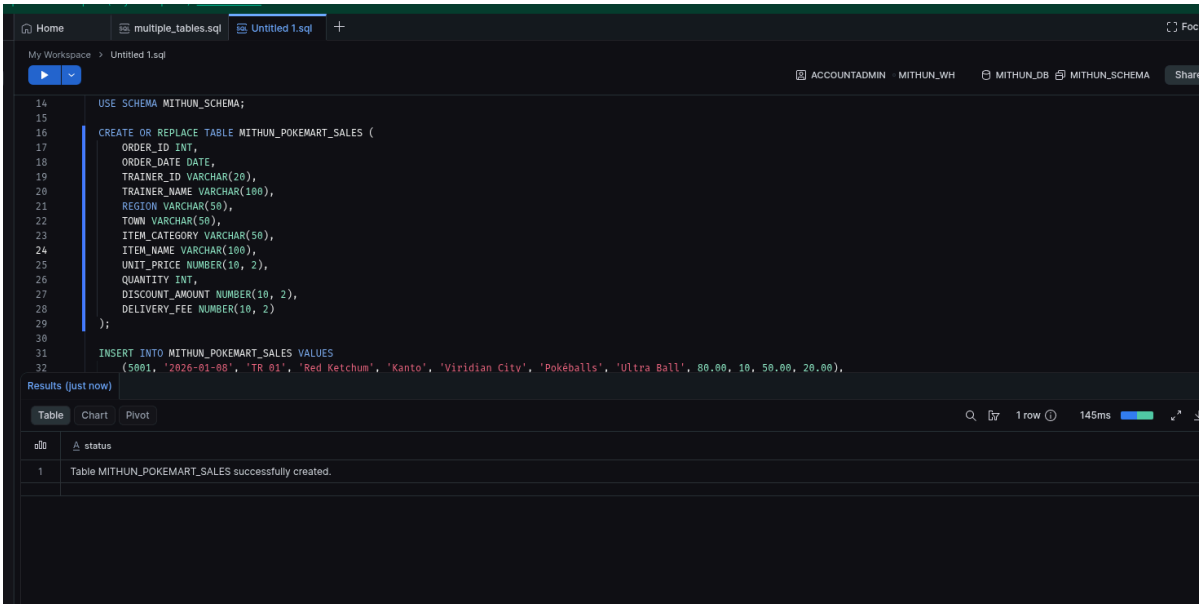
Snowflake to Power BI Integration & Dashboard Creation

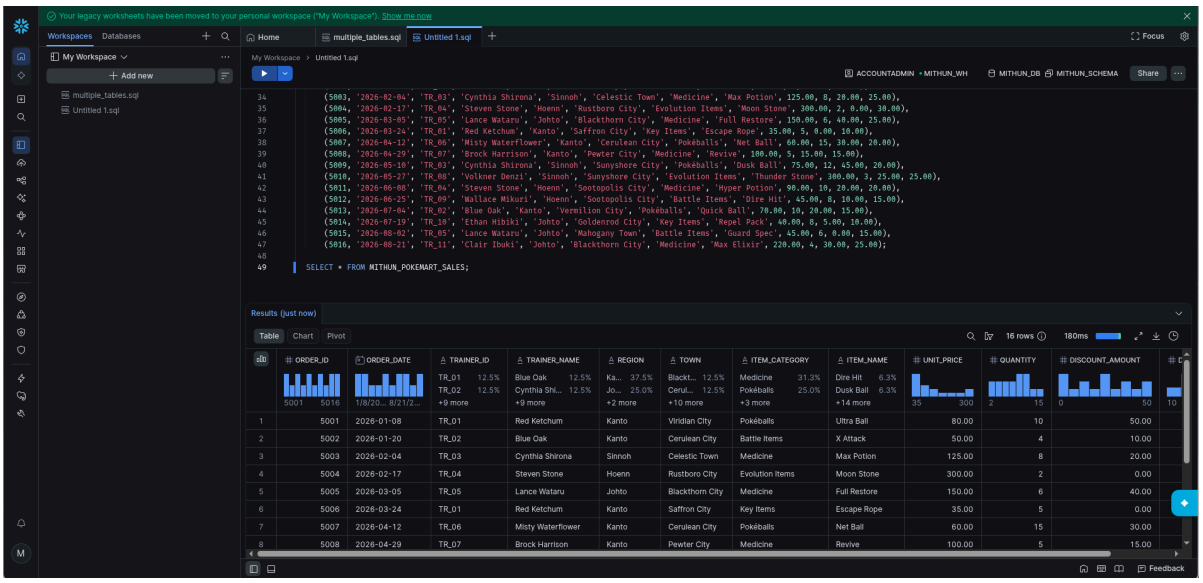
Step 1: Snowflake Environment Setup & Data Ingestion

Objective: Initialize a dedicated virtual warehouse, database, and schema in Snowflake, and populate a raw table with sample transaction data.

Description:

1. Open a new SQL Worksheet in the Snowflake UI.
2. Execute the setup script to provision the infrastructure and load the data.
3. Verify the data using a SELECT query.
4. Note down the warehouse name (MITHUN_WH) and Server URL.



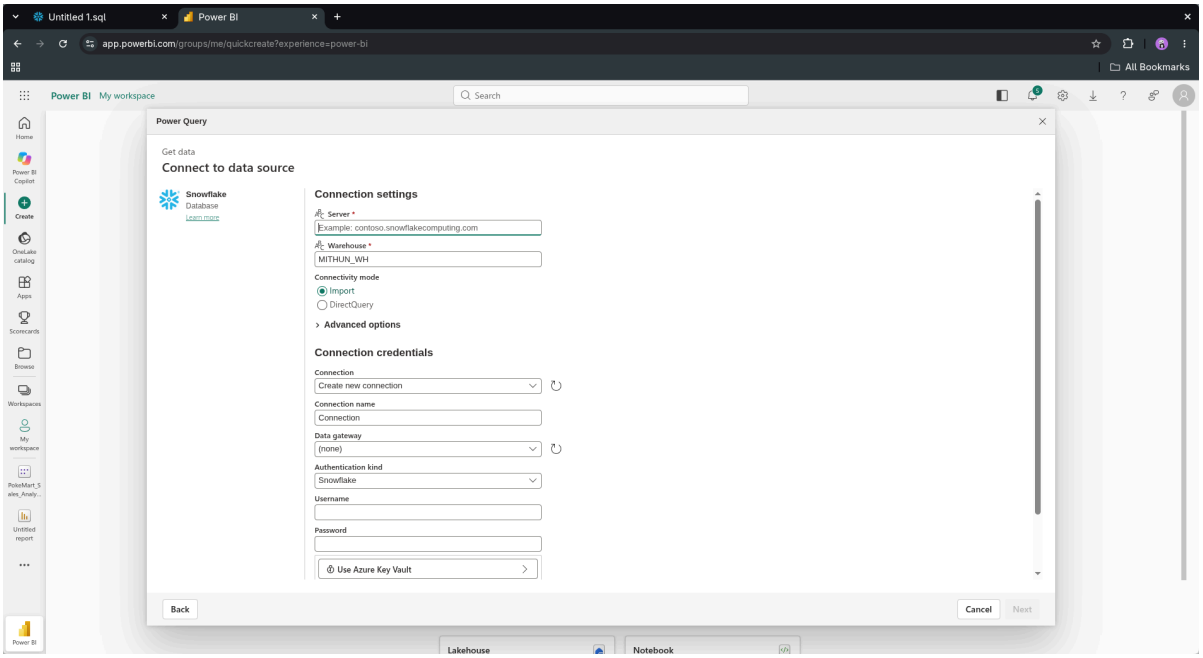


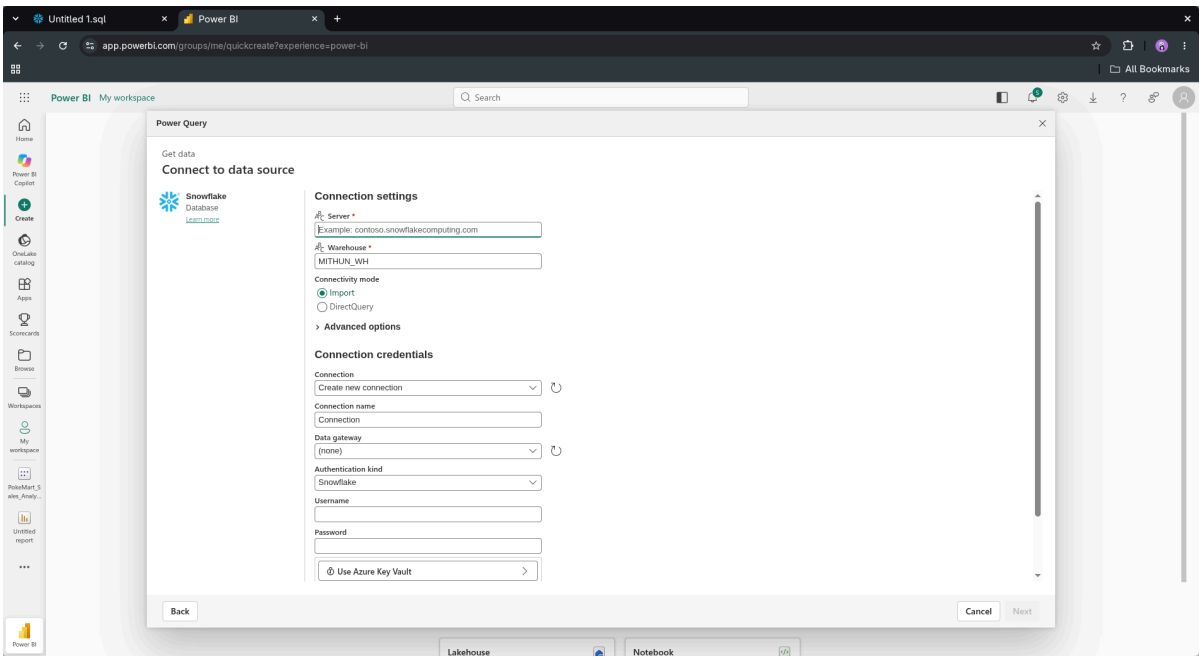
Step 2: Establish Power BI Connection

Objective: Securely connect Power BI Desktop to the Snowflake data warehouse.

Description:

1. Open Power BI Desktop and select Get Data > Snowflake.
2. Input the Server URL and Warehouse name (MITHUN_WH).
3. Authenticate and select the MITHUN_POKEMART_SALES table.





Step 3: Create DAX Calculated Columns

Objective: Create row-level calculated columns in Power BI to derive sales revenue, estimated product cost, gross profit, and profit classification for each transaction.
Before starting: Open the model in Editing mode, select the **pokemart_sales 1** table, and choose Table view or Model view. Use Table tools / Modeling > New column. Create each formula separately.

3.1 Sales Amount

Objective: Calculate the revenue from each sale after subtracting the discount. Tax is excluded from revenue.
Column name: Sales Amount

```
Sales Amount =
('pokemart_sales 1'[Quantity] * 'pokemart_sales 1'[UnitPrice])
- 'pokemart_sales 1'[Discount]
```

Explanation: Quantity multiplied by UnitPrice gives the gross item value. Discount is then subtracted to calculate net sales revenue for the transaction. Tax is not included because it is collected separately from revenue.
[Insert screenshot: Sales Amount calculated column and its values.]

3.2 Total Cost

Objective: Estimate the cost of products sold for each transaction. Actual product cost is not present in the supplied dataset, so product cost is assumed to be 70% of Sales Amount.

Column name: Total Cost

```
Total Cost =
'pokemart_sales 1'[Sales Amount] * 0.70
```

Explanation: This calculated column applies the 70% cost assumption to the net Sales Amount of each sale. It is an estimate, not an actual supplier or inventory cost.
[Insert screenshot: Total Cost calculated column and its values.]

3.3 Gross Profit

Objective: Calculate the estimated profit earned on each transaction after deducting product cost.
Column name: Gross Profit

```
Gross Profit =
'pokemart_sales 1'[Sales Amount] - 'pokemart_sales 1'[Total Cost]
```

Explanation: Gross Profit is calculated by subtracting the estimated Total Cost from Sales Amount.
[Insert screenshot: Gross Profit calculated column and its values.]

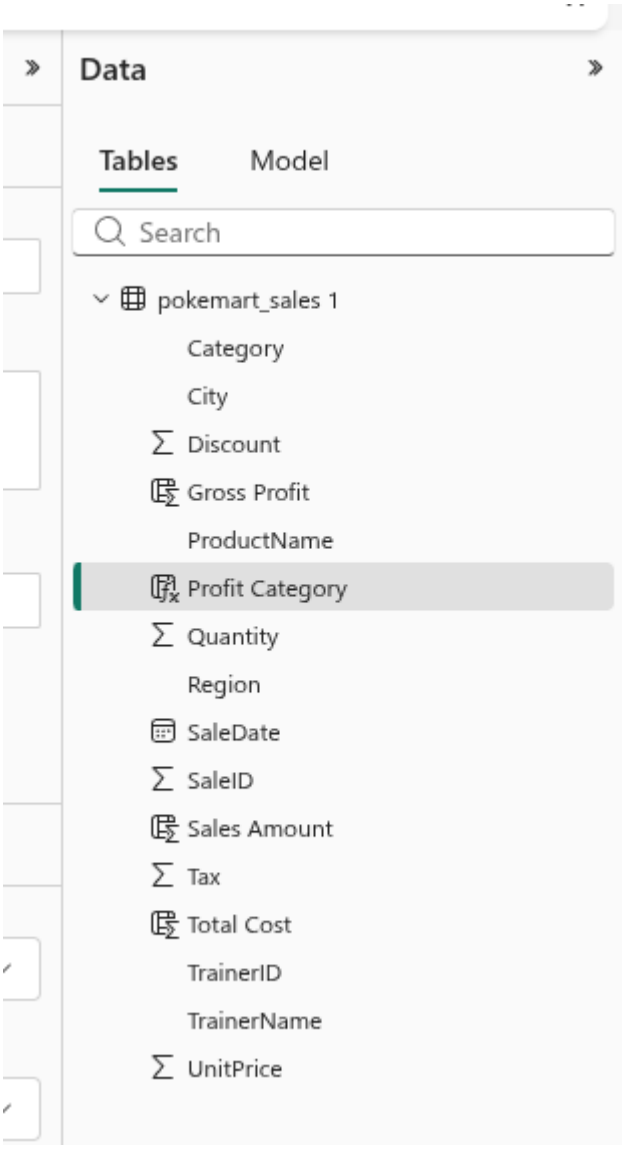
3.4 Profit Category

Objective: Classify each transaction according to its estimated gross profit.
Column name: Profit Category

```
Profit Category =
SWITCH(
    TRUE(),
    'pokemart_sales 1'[Gross Profit] >= 1000, "Higher Profit",
    'pokemart_sales 1'[Gross Profit] >= 300, "Medium Profit",
    "Lower Profit"
)
```

Explanation: SWITCH(TRUE()) checks the conditions from top to bottom and assigns a profit label to each sale. The thresholds are scaled for the price range of the supplied PokéMart sample data.
[Insert screenshot: Profit Category calculated column and its values.]





Step 4: Create DAX Measures

Objective: Create DAX measures for business-level sales, profitability, customer, order, and target performance metrics. Measures respond dynamically to report filters.

Procedure: Select the pokemart_sales 1 table in the Data pane, choose New measure, enter one formula at a time, and press Enter.

4.1 Total Sales

```
Total Sales =
SUM('pokemart_sales 1'[Sales Amount])
```

Explanation: Adds Sales Amount for the current filter context.

4.2 Total Cost

```
Total Cost =
SUM('pokemart_sales 1'[Total Cost])
```

Explanation: Adds estimated product cost for the current filter context.

4.3 Total Gross Profit

```
Total Gross Profit =
SUM('pokemart_sales 1'[Gross Profit])
```

Explanation: Adds gross profit for the current filter context.

4.4 Total Margin

```
Total Margin =
DIVIDE([Total Gross Profit], [Total Sales], 0)
```

Explanation: Calculates the gross-profit-to-sales ratio as a decimal.

4.5 Profit Margin

```
Profit Margin =
DIVIDE([Total Gross Profit], [Total Sales], 0) * 100
```

Explanation: Calculates gross profit as a percentage of sales. Since the formula multiplies by 100, format it as a decimal number, not Power BI Percentage.

4.6 Total Customers

```
Total Customers =
DISTINCTCOUNT('pokemart_sales 1'[TrainerID])
```

Explanation: Counts unique TrainerID values, treating trainers as customers for this sample analysis.

4.7 Total Orders

```
Total Orders =
DISTINCTCOUNT('pokemart_sales 1'[SaleID])
```

Explanation: Counts distinct sale identifiers.

4.8 Average Order Value

```
Average Order Value =
DIVIDE([Total Sales], [Total Orders], 0)
```

Explanation: Calculates average sales revenue per order.

4.9 Sales Target

```
Sales Target = 100000
```

Explanation: Defines the fixed sales target as ₹1,00,000.

4.10 Target Achievement

```
Target Achievement =
DIVIDE([Total Sales], [Sales Target], 0) * 100
```

Explanation: Calculates the percentage of target achieved. Since the formula multiplies by 100, format it as a decimal number.

4.11 Target Not Achieved %

```
Target Not Achieved % =
MAX(0, 100 - [Target Achievement])
```

Explanation: Calculates the remaining percentage to reach the target, with a minimum of zero.

4.12 Target Shortfall

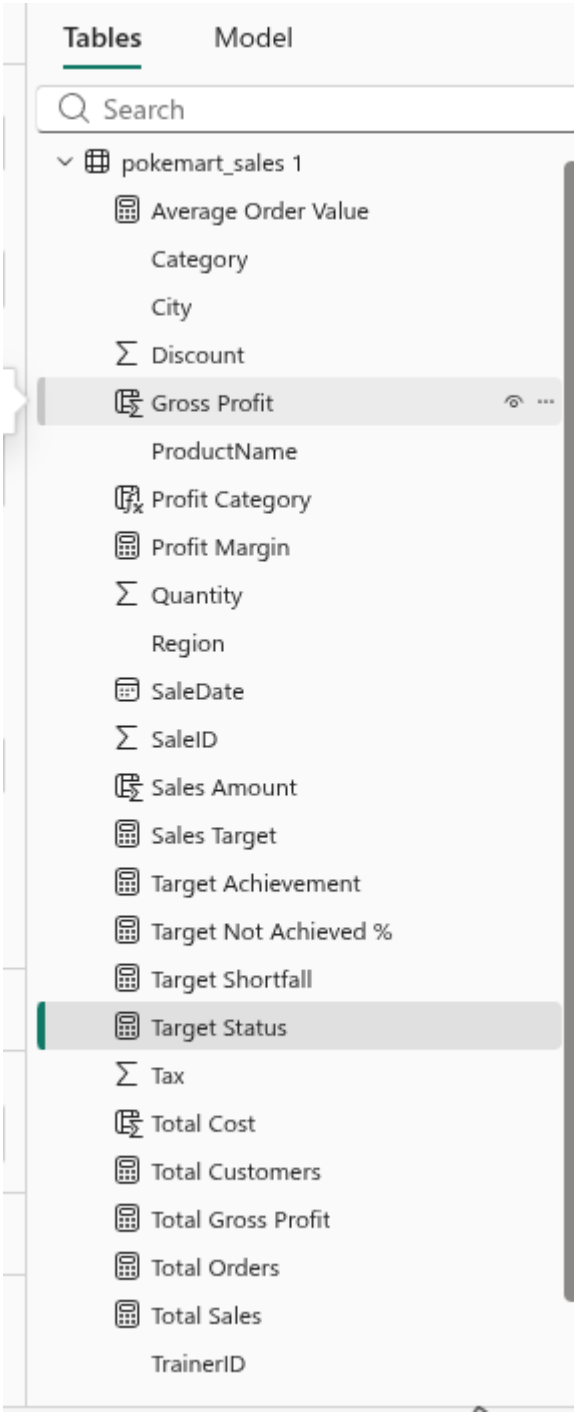
```
Target Shortfall =
MAX(0, [Sales Target] - [Total Sales])
```

Explanation: Calculates the amount still required to reach the target, with a minimum of zero.

4.13 Target Status

```
Target Status =
IF(
    [Total Sales] >= [Sales Target],
    "Target Achieved",
    "Target Not Achieved"
)
```

Explanation: Displays whether total sales have reached the predefined target.


Tables
Model
Search
pokemart_sales 1
Average Order Value
Category
City
Discount
Gross Profit
ProductName
Profit Category
Profit Margin
Quantity
Region
SaleDate
SaleID
Sales Amount
Sales Target
Target Achievement
Target Not Achieved %
Target Shortfall
Target Status
Tax
Total Cost
Total Customers
Total Gross Profit
Total Orders
Total Sales
TrainerID

Step 5: Build the Sales Performance Dashboard

